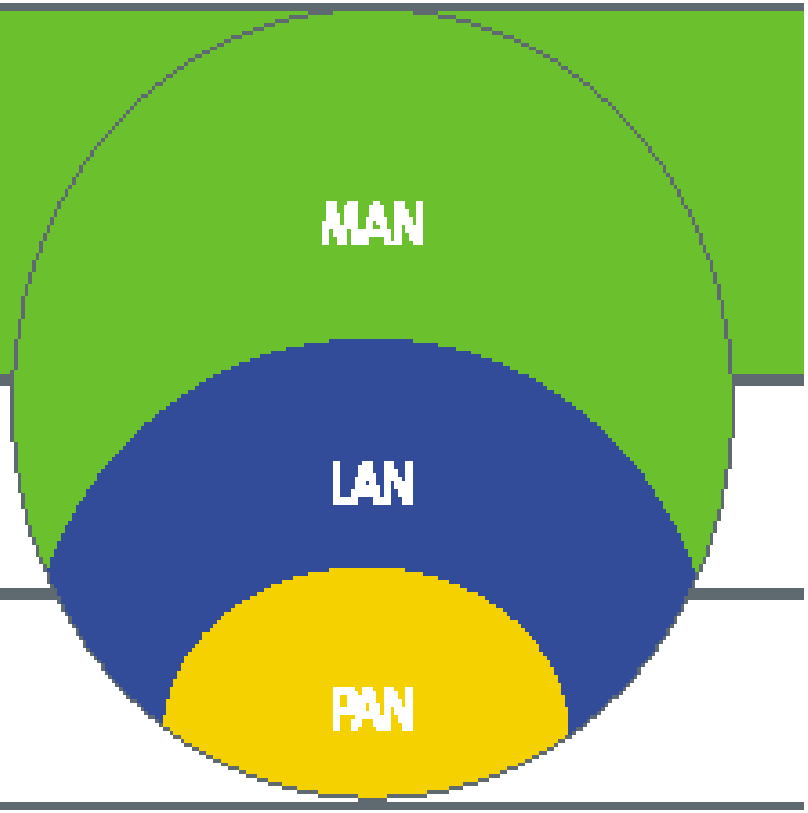
Simulation of Wireless Networks

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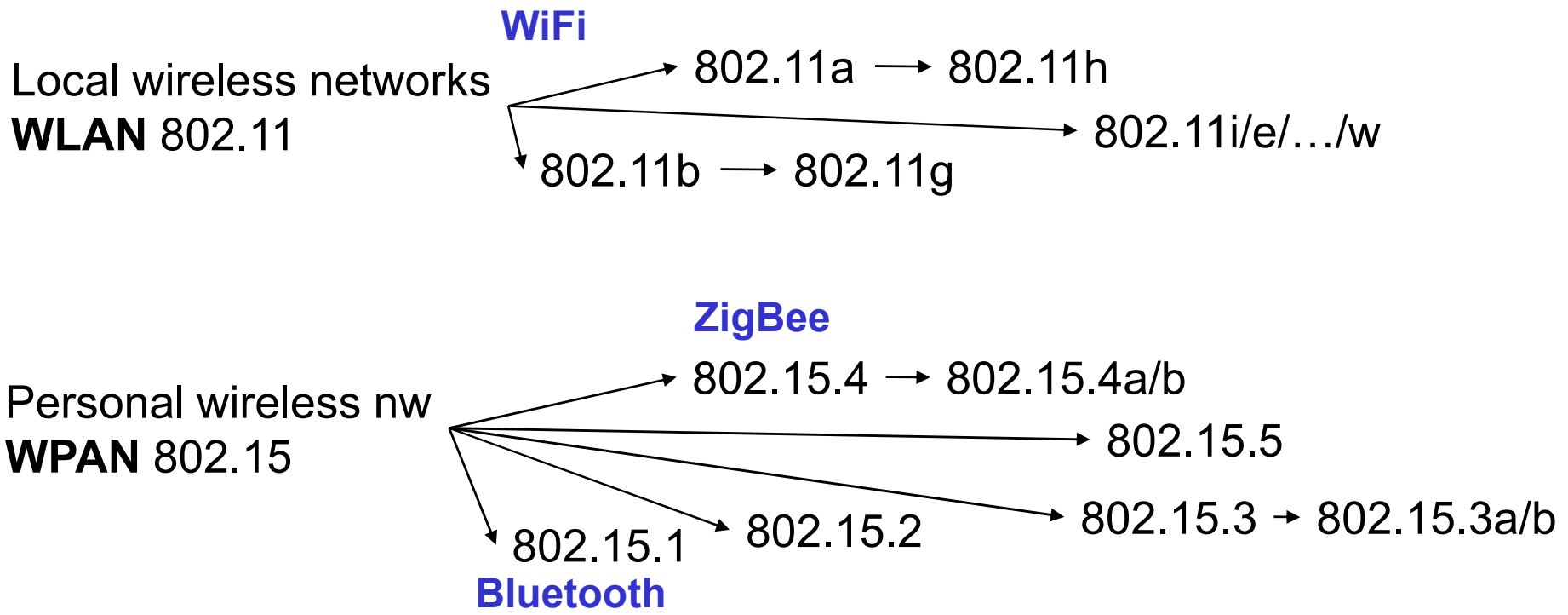
Wireless Networks

WiMAX	802.16/HiperMAN
WiFi	802.11
Bluetooth	802.15



Wireless standards and their networking environments

Wireless Networks according to IEEE



Wireless distribution networks
WMAN 802.16 (Broadband Wireless Access) **WiMAX**

+ Mobility

802.20 (Mobile Broadband Wireless Access)

Applications of Wireless Networks

- Wireless PANs (Bluetooth – IEEE 802.15)
  - very low range
  - wireless connection to printers etc
- Wireless LANs (WiFi – IEEE 802.11)
  - infrastructure as well as ad-hoc networks possible
  - home/office networking
- Multihop Ad hoc Networks
  - useful when infrastructure not available, impractical, or expensive
  - military applications, emergencies
- Wireless MANs (WiMAX-802.16)
  - Similar to cellular networks
  - traditional base station infrastructure systems



Application of Wi-Fi

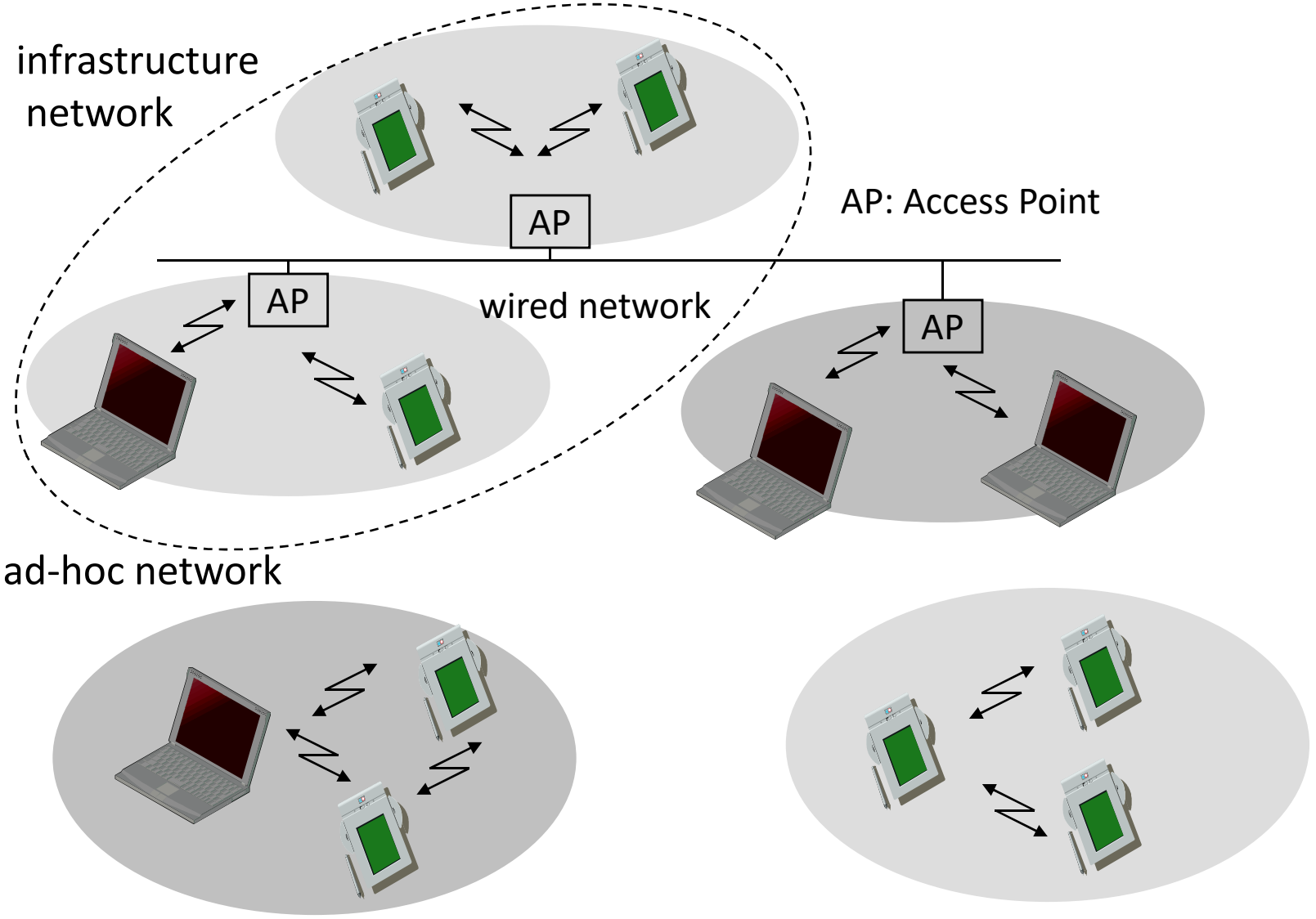


- Many electronic devices use Wi-Fi due to its simple functions.
- Companies use Wi-Fi to create wireless networks within their company.
- Phone companies use Wi-Fi Hot Spots for their users to get free calling and internet access.
- Gaming companies like Nintendo use Wi-Fi to synch their products and to use global network usage.

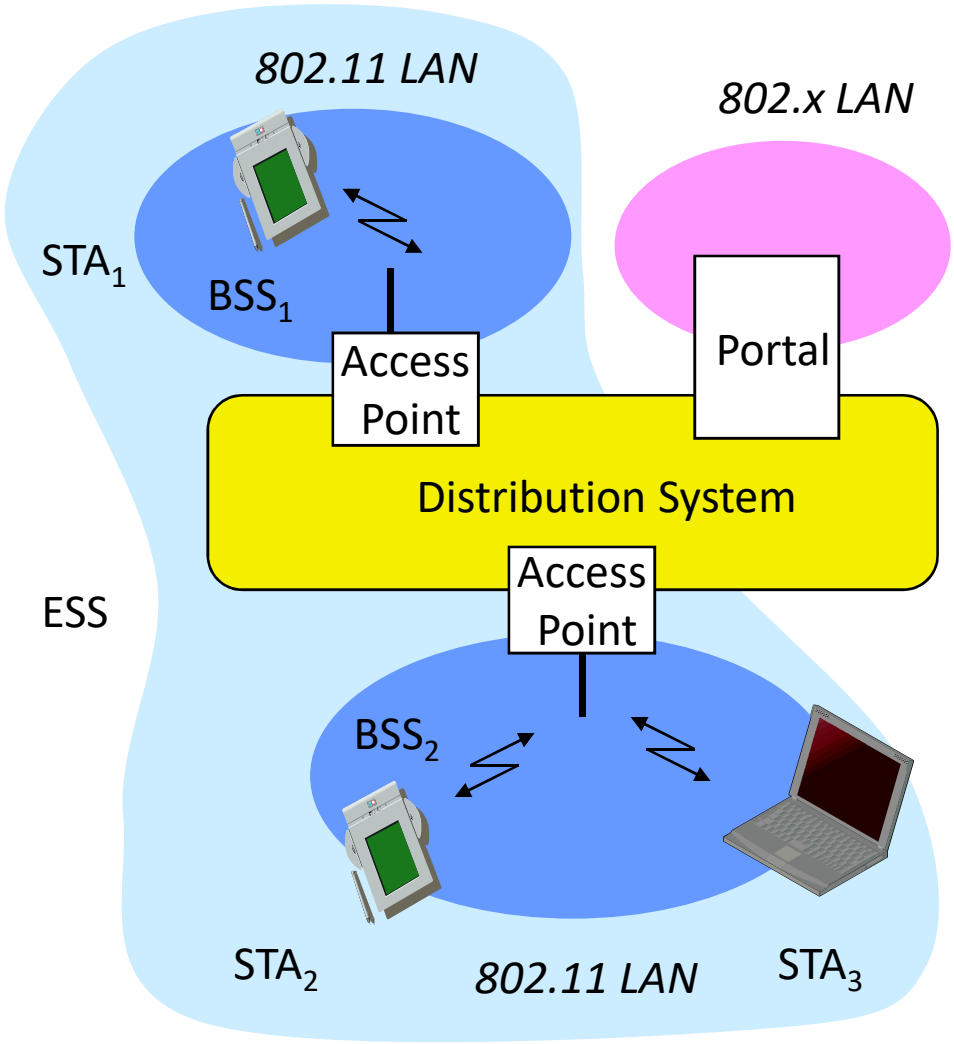




WiFi: infrastructure vs. ad-hoc networks

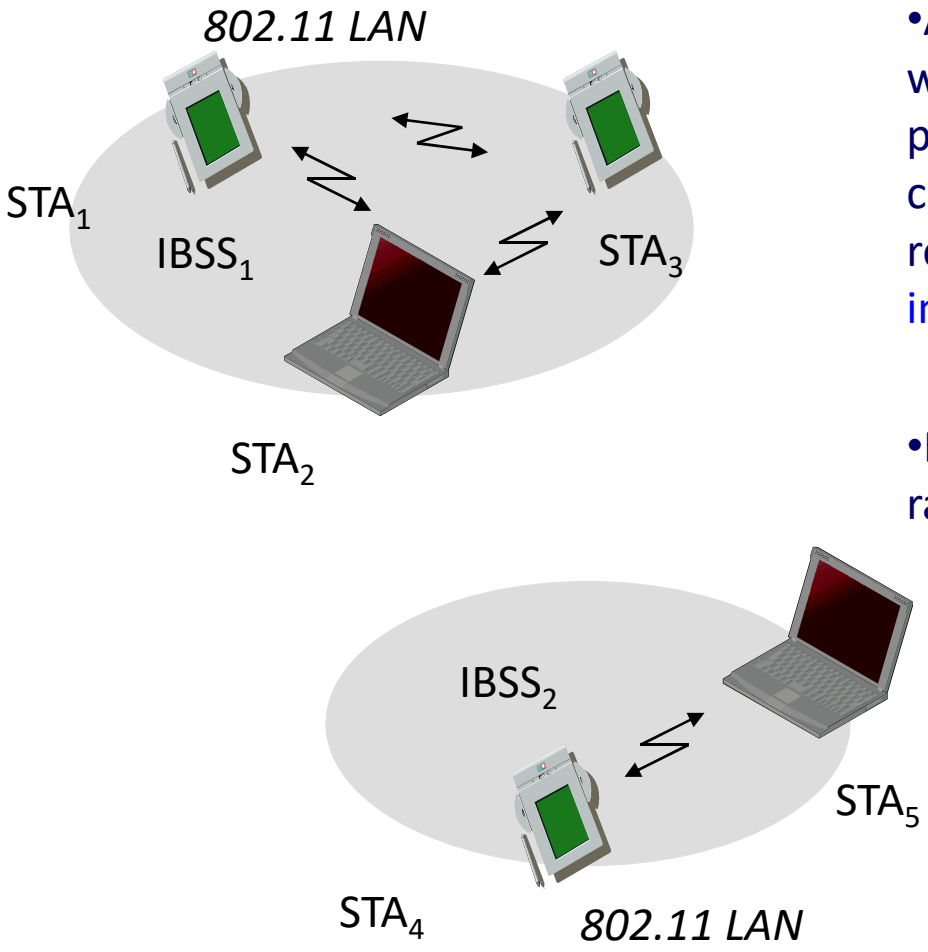


802.11 - Infrastructure Network Architecture



- Station (STA)
  - terminal with access mechanisms to the wireless medium and radio contact to the access point
- Basic Service Set (BSS)
  - group of stations using the same radio frequency
- Access Point
  - station integrated into the wireless LAN and the distribution system
- Portal
  - bridge to other (wired) networks
- Distribution System
  - interconnection network to form one logical network (EES: Extended Service Set) based on several BSS

802.11 – Ad Hoc Network Architecture



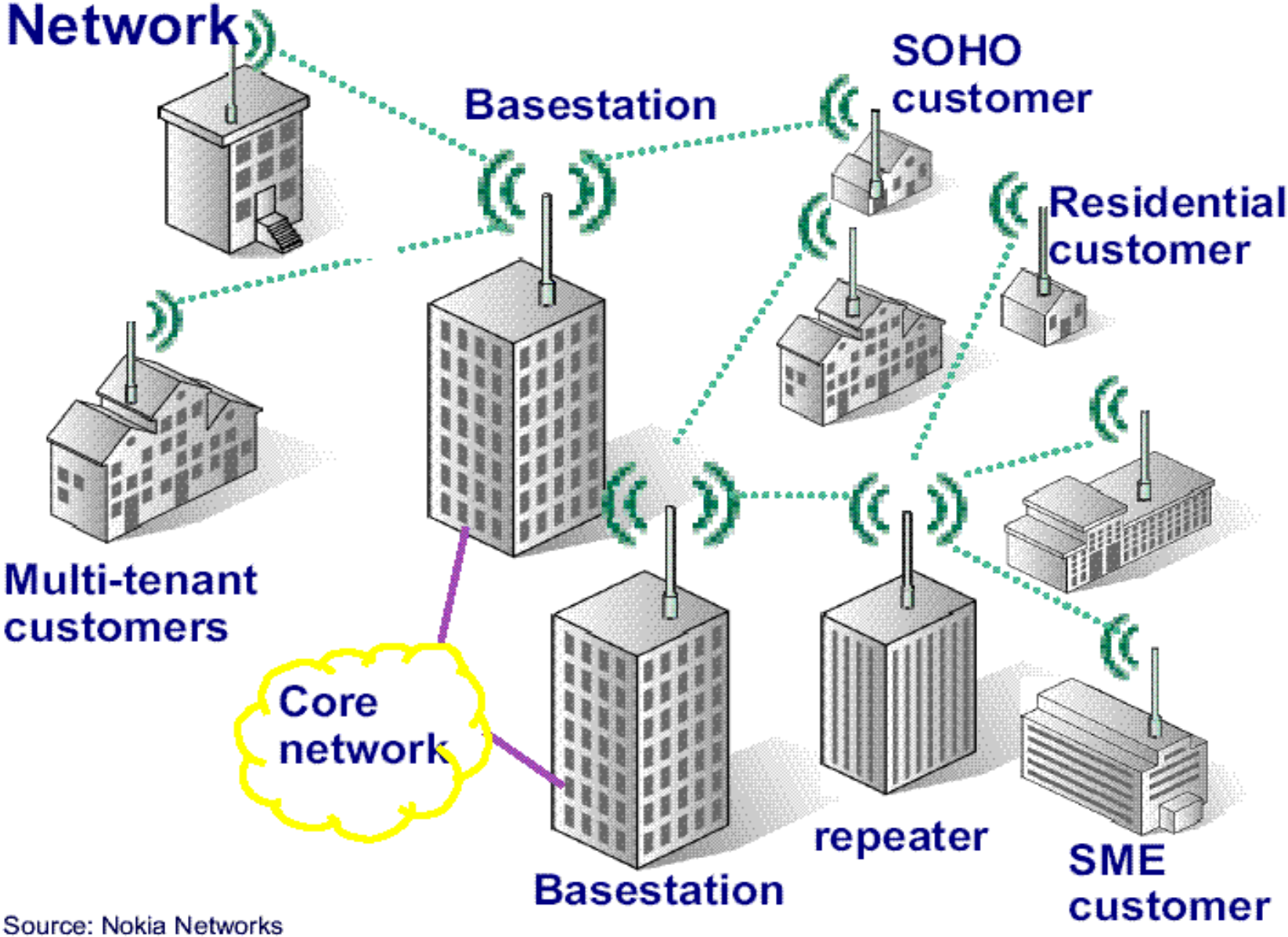
- An ad hoc network is a self-configurable wireless network in which mobile, battery powered devices dynamically create and change the network topology without relying on any fixed networking infrastructure or administrative support.

- Direct communication within a limited range
  - Station (STA): terminal with access mechanisms to the wireless medium
  - Independent Basic Service Set (IBSS): group of stations using the same radio frequency

WiMAX – Broadband Wireless Access

- Goal: Provide high-speed Internet access to home and business subscribers, without wires.
- Base stations (BS) and subscriber stations (SS)
- Centralized access control to prevents collisions
- Supports applications with different QoS requirements
- WiMAX is a subset of IEEE 802.16 standard

WirelessMAN: Wireless Metropolitan Area Network



Source: Nokia Networks

IEEE 802.16 Standards

- 802.16.1 (10-66 GHz, line-of-sight, up to 134Mbit/s)
- 802.16.2 (minimizing interference between coexisting WMANs)

- 802.16a (2-11 Ghz, Mesh, non-line-of-sight)
- 802.16b (5-6 Ghz)
- 802.16c (detailed system profiles)

- 802.16e (Mobile Wireless MAN)

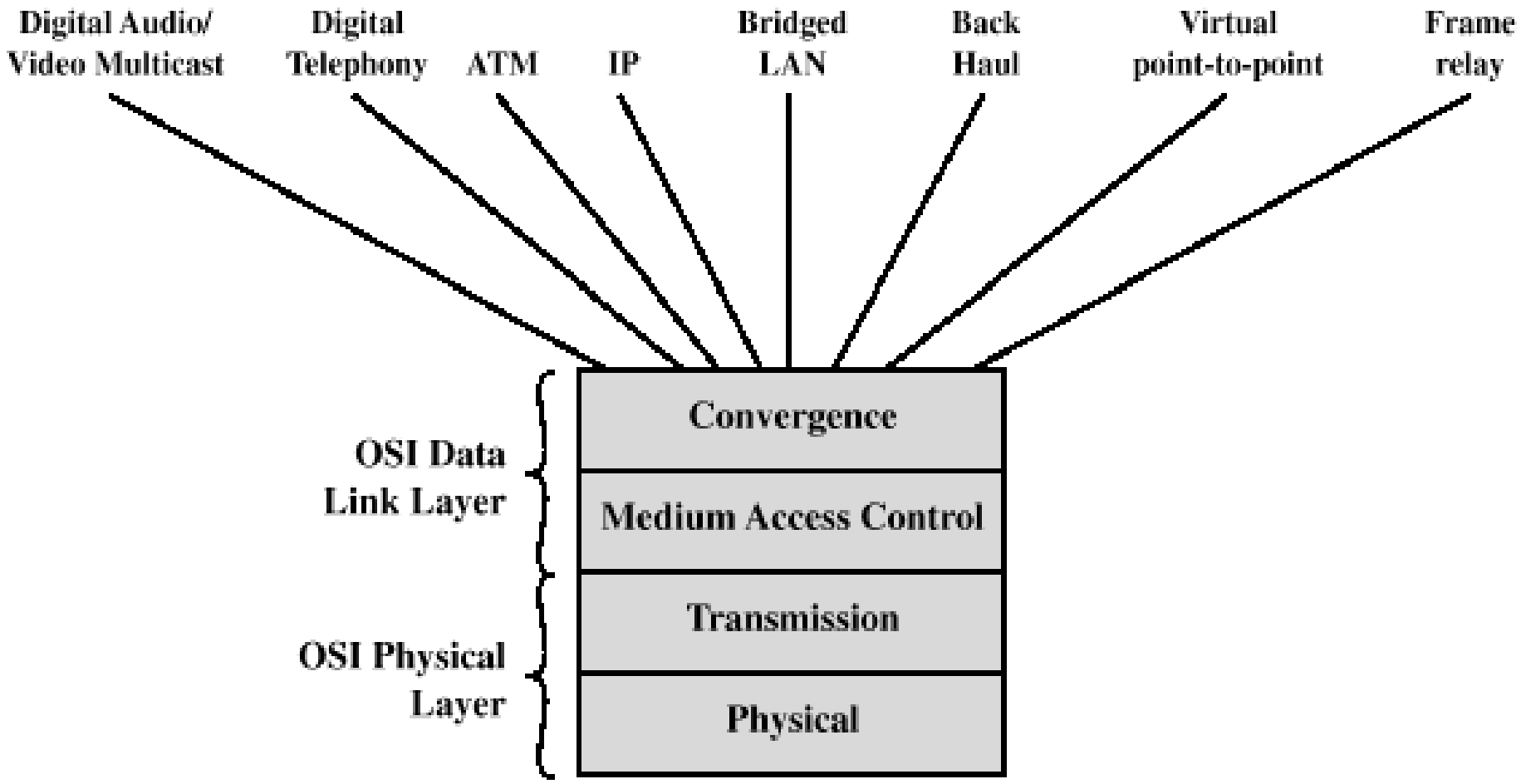
802.16 Standards

- Use wireless links with microwave or millimeter wave radios
  - 10-66 GHz
  - 802.16a extension to 2-11 GHz
- Use licensed spectrum (unlicensed too in 802.16a)
- Metropolitan in scale
- Provide public network service to fee-paying customers
- Point-to-multipoint architecture with rooftop or tower-mounted antennas

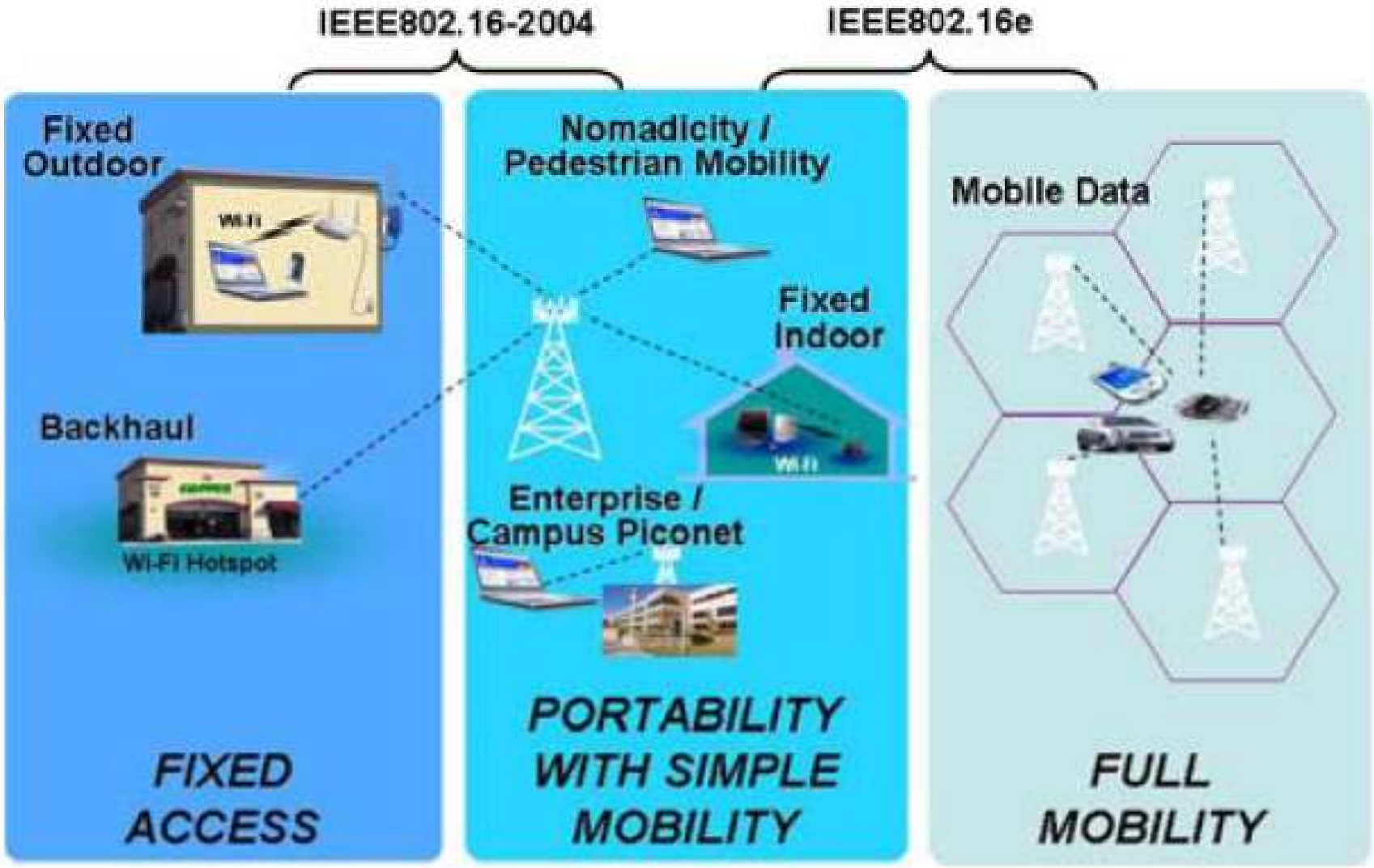
802.16 Standards (Contd..)

- Provide efficient transport of heterogeneous traffic supporting QoS
- Capable of broadband transmissions (2-75 Mbps)
  - Accommodate both continuous and bursty traffic
- Mobile extensions: 802.16e

IEEE 802.16 Protocol Architecture



Applications of 802.16 Stds



Parts of a ns3 Program

- Header Part // #include “ns3/XXX-module.h”
- Node Creation
- Topology Creation
- Assign Devices
- Install Protocol Stack
- Assign Address – supports both IPv4 & IPv6
- Write code for Application Part
- Enable Log Components
- Turn on tracing

/* Sample ns3 code – first.cc */

```
#include "ns3/core-module.h"
#include "ns3/network-module.h"
#include "ns3/internet-module.h"
#include "ns3/point-to-point-module.h"
#include "ns3/applications-module.h"

using namespace ns3;

NS_LOG_COMPONENT_DEFINE ("FirstScriptExample");

int main (int argc, char *argv[]) {
  NodeContainer nodes;
  nodes.Create (2);

[…]

  ApplicationContainer clientApps = echoClient.Install
(nodes.Get (0));
  clientApps.Start (Seconds (2.0));
  clientApps.Stop (Seconds (10.0));

  Simulator::Run ();
  Simulator::Destroy ();
  return 0;
}
```

Compile / Run ns3 Programs

- ./waf – main command in ns3
- Open Terminal -> Run ns3 programs in the ns-allinone-3.24.1/ns-3.24.1 folder
- Copy any program to be run into the ns-allinone-3.24.1/ns-3.24.1/scratch folder
- Only the files in the scratch folder are compiled by the ./waf command

View Animation using NetAnim

- Go to ns-allinone-3.24.1/netanim-3.106 folder
- Run/Open (click on) NetAnim.exe file
- Click on the “Open XML trace file” icon and select filename.xml
- Click on the Run button to view animation

Hands-on Network Simulation in ns3

- Open Terminal and in the command prompt ns3@ubuntu:~/ns-allinone-3.24.1/ns-3.24.1$
- Copy file to run from its folder to scratch
  - cp src/netanim/examples/wireless-animation.cc scratch/wirelessanim.cc
- gedit scratch/wirelessanim.cc
- Add #include “ns3/netanim-module.h” in the header part of the code

Hands-on Network Simulation (contd…)

- Before Simulator::Run(), add AnimationInterface anim(“filename.xml”);
- Save File (click on save icon)
- In command prompt of Terminal, that is ns3@ubuntu:~/ns-allinone-3.24.1/ns-3.24.1$ ./waf <Enter>
- ./waf –run wirelessanim <Enter>
- Only “wirelessanim” & not “wirelessanim.cc”

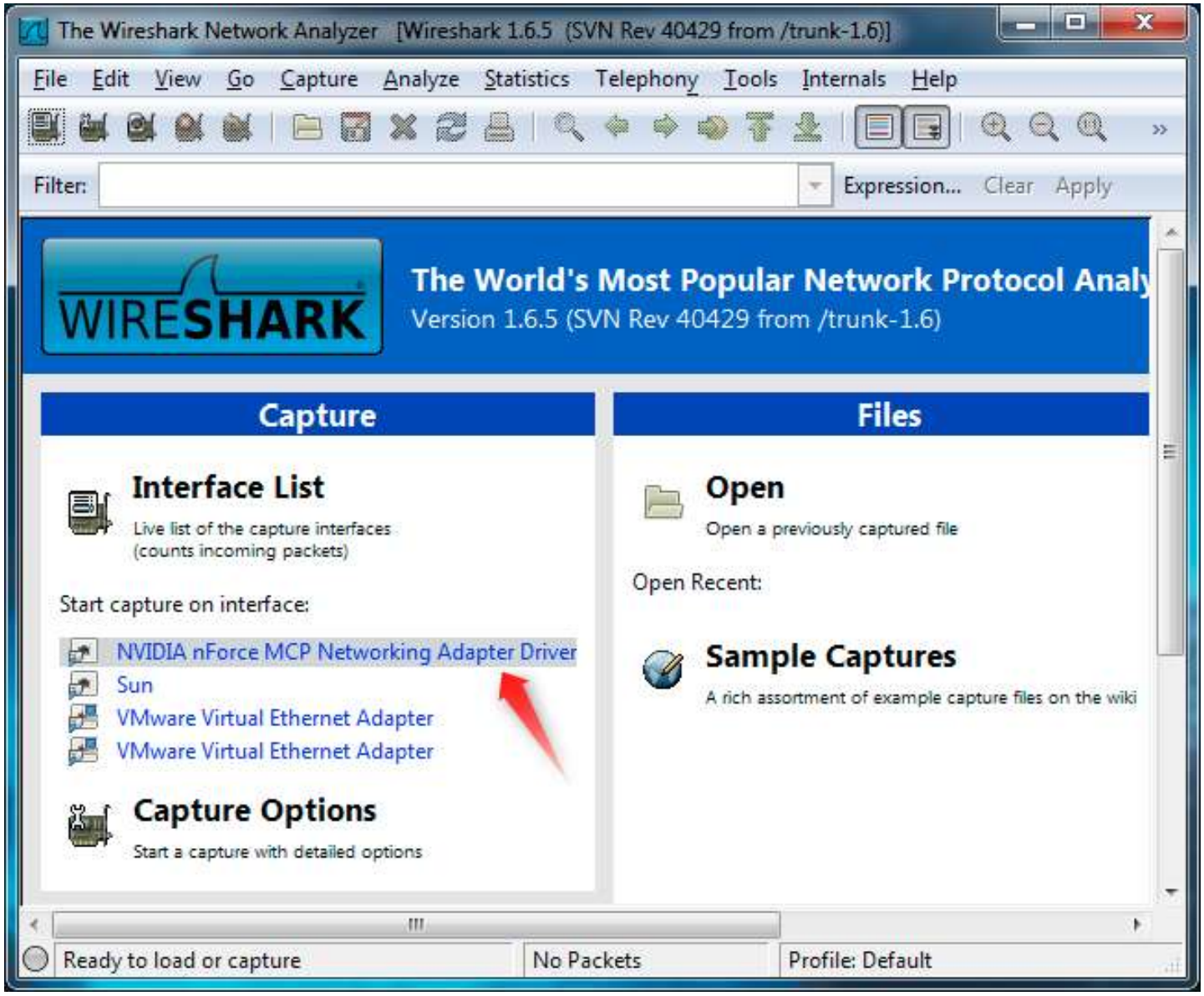
Hands-on Network Simulation (contd…)

- ns-allinone-3.24.1/ns-3.24.1/src/wimax/examples/wimax-simple.cc
- ns-allinone-3.24.1/ns-3.24.1/src/wimax/examples/wimax-ipv4.cc
- ns-allinone-3.24.1/ns-3.24.1/src/wimax/examples/wimax-multicast.cc
- ns-allinone-3.24.1/ns-3.24.1/examples/tutorial/third.cc

Hands-on Network Simulation (contd…)

- ns-allinone-3.24.1/ns-3.24.1/src/lte/examples/lena-pathloss-traces.cc
- ns-allinone-3.24.1/ns-3.24.1/src/lte/examples/lena-x2-handover.cc
- ns-allinone-3.24.1/ns-3.24.1/examples/routing/examples/manet-routing-compare.cc

Wireshark


The Wireshark Network Analyzer [Wireshark 1.6.5 (SVN Rev 40429 from /trunk-1.6)]
File Edit View Go Capture Analyze Statistics Telephony Tools Internals Help
Filter:
Expression... Clear Apply
WIRESHARK
The World's Most Popular Network Protocol Analy
Version 1.6.5 (SVN Rev 40429 from /trunk-1.6)
Capture
Interface List
Live list of the capture interfaces
(counts incoming packets)
Start capture on interface:
NVIDIA nForce MCP Networking Adapter Driver
Sun
VMware Virtual Ethernet Adapter
VMware Virtual Ethernet Adapter
Capture Options
Start a capture with detailed options
Files
Open
Open a previously captured file
Open Recent:
Sample Captures
A rich assortment of example capture files on the wiki
Ready to load or capture
No Packets
Profile: Default

Wireshark (Contd…)

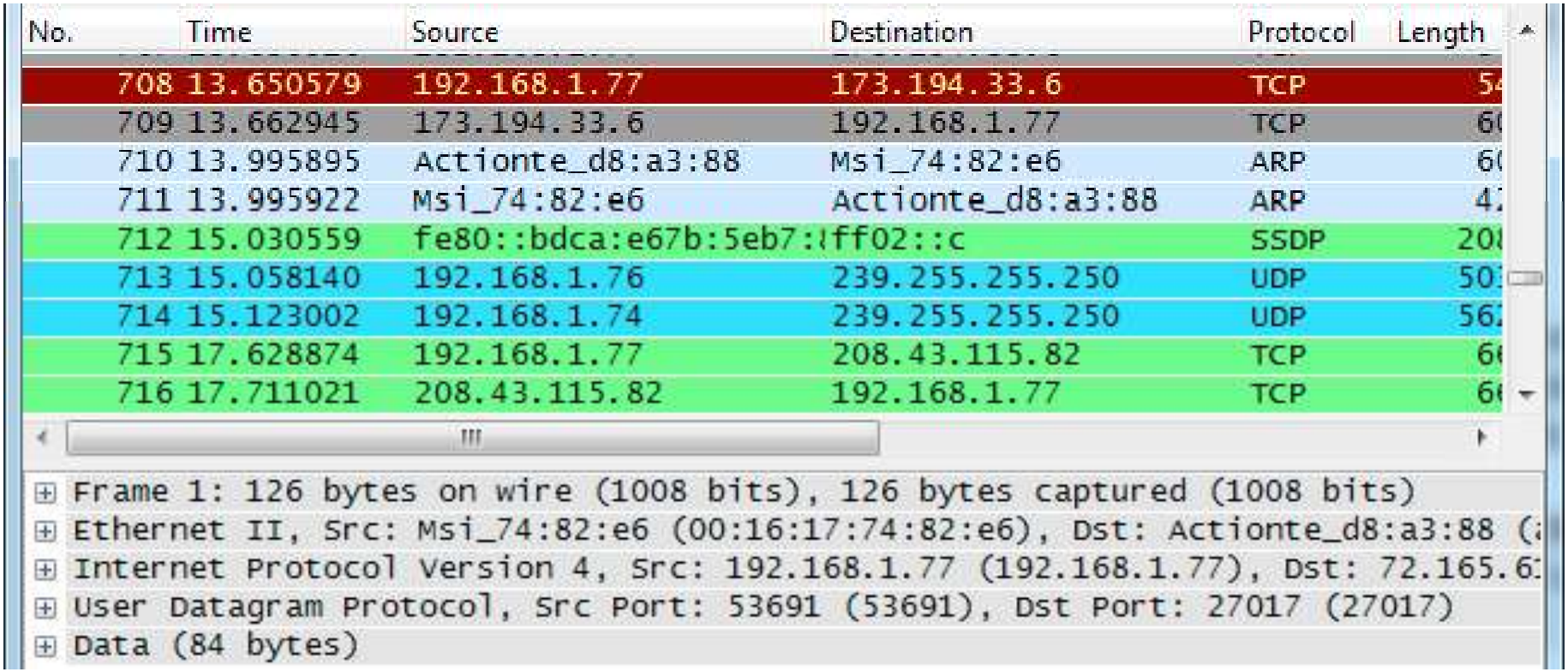
- World's foremost network protocol analyzer
- The de facto standard across many industries and educational institutions
- Deep inspection of hundreds of protocols, with more being added all the time
- Live capture and offline analysis
- Standard three-pane packet browser
- Multi-platform: Runs on Windows, Linux, OS X, Solaris, FreeBSD, NetBSD, and many others
- Captured network data can be browsed via a GUI, or via the TTY-mode TShark utility
- The most powerful display filters in the industry

Wireshark (Contd…)

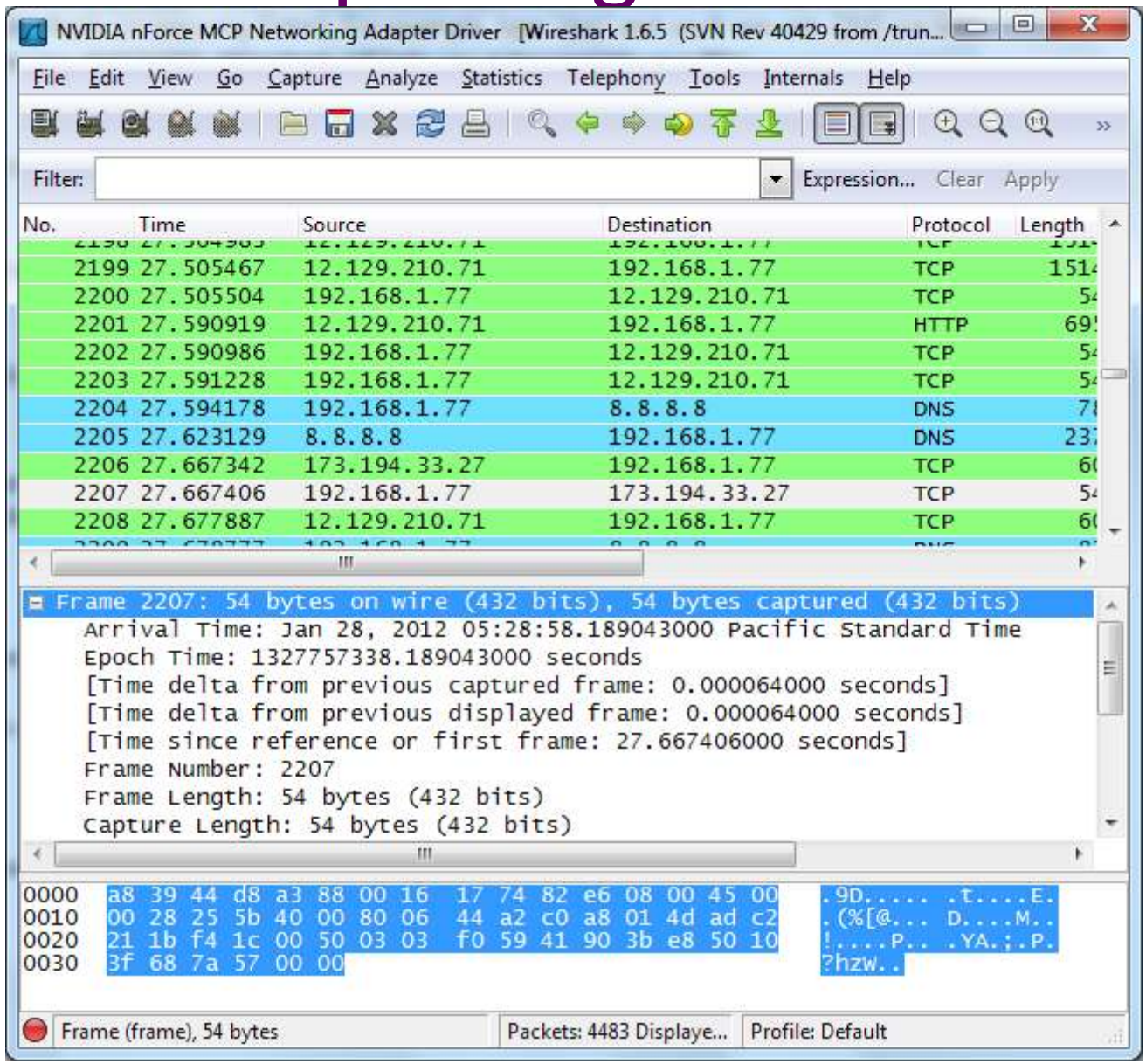
- Rich VoIP analysis
- Read/write many different capture file formats
- Capture files compressed with gzip can be decompressed on the fly
- Live data can be read from Ethernet, IEEE 802.11, PPP/HDLC, ATM, Bluetooth, USB, Token Ring, Frame Relay, FDDI, and others (depending on your platfrom)
- Decryption support for many protocols, including IPsec, ISAKMP, Kerberos, SNMPv3, SSL/TLS, WEP, and WPA/WPA2
- Coloring rules can be applied to the packet list for quick, intuitive analysis
- Output can be exported to XML, PostScript®, CSV, or plain text

Wireshark (Contd…)

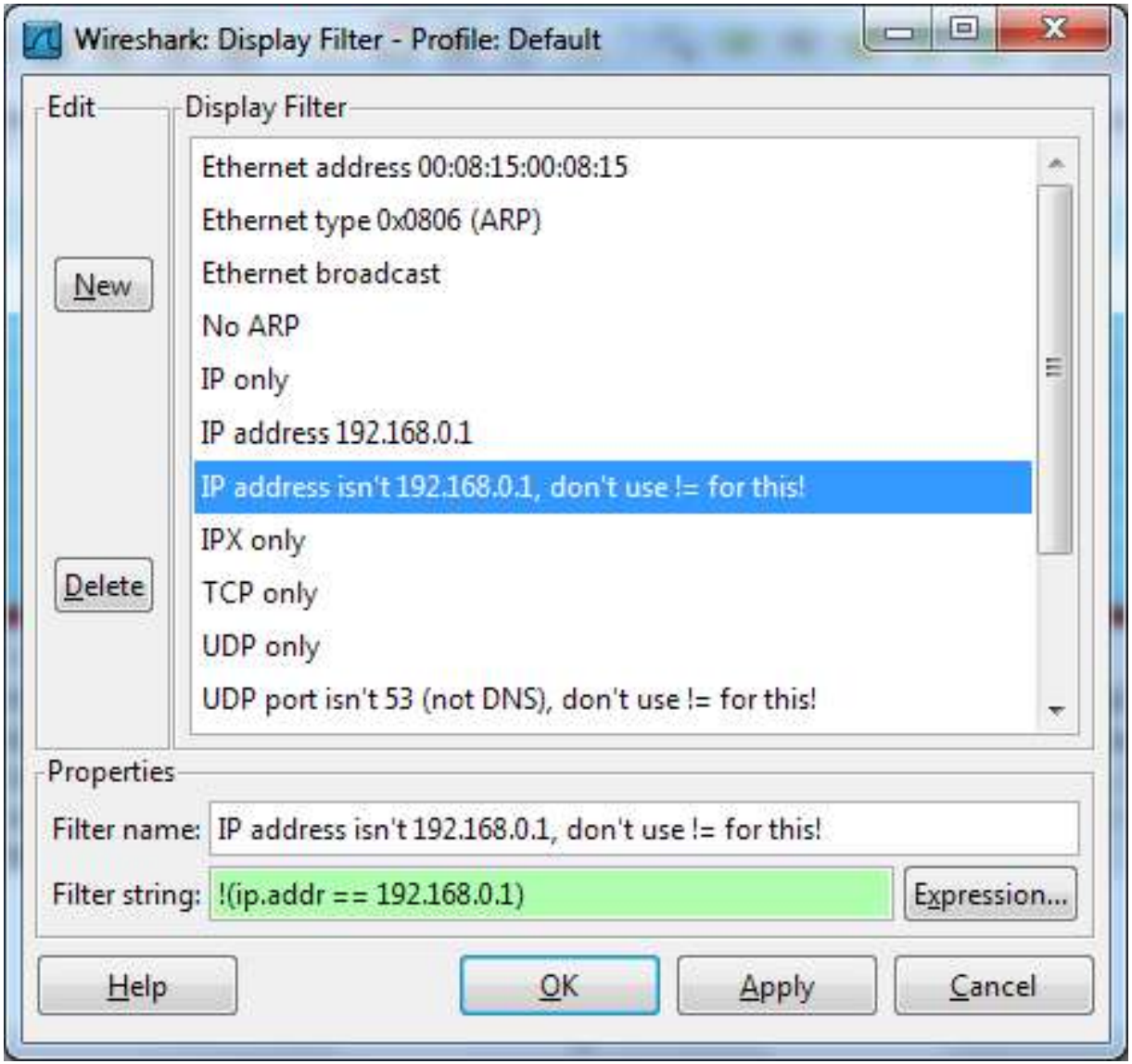
Capture, Filter and Inspect Packets:



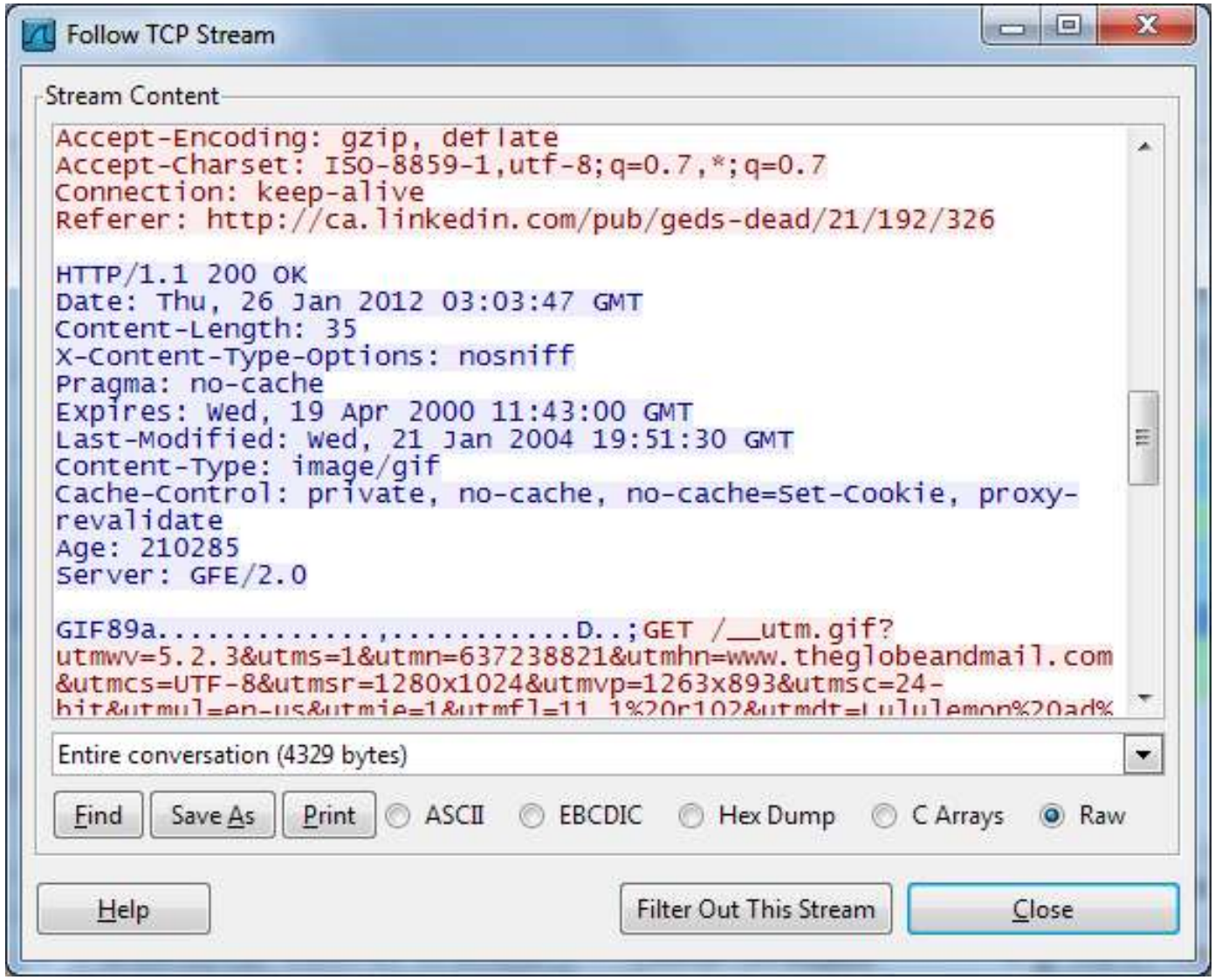
Inspecting Packets



Wireshark – Display Filter



Wireshark – Follow TCP Stream



Adding a new Program

- src is the core folder in ns3
- To add anew program inside src or any folder inside src:
- Go to src/wimax/examples: copy wimax-simple.ccand paste it in the same folder and renameit as wimax-simple2.cc
- Now, add following header file, in the new file:

  #include “ns3/netanim-module.h”

Adding a new Program (Contd…)

- Copy the new file to scratch folder and run
- Compiler displays error message: “No such file exist”
- To avoid above error, open the wscript file in the folder of the new file using gedit and add the following:

```
obj = bld.create_ns3_program( 'wimax-simple2',
    [‘netanim’, ‘wimax', 'internet', 'mobility',
    'applications'])
obj.source = 'wimax-simple.cc'
```

Enjoy

Develop exciting network simulations using ns3 and contribute/share your ideas/experiences with the Academia and Research Community

THANK YOU